

The Naylor Blueprint

A Full-Spectrum Basestealing Leaderboard

ONE leaderboard, two poles — ranked by a single number.

TOP — the blueprint:
the SLOWEST runners who cover the most ground between the pitcher's first move and release, and convert it into steals.

BOTTOM — the anti-Naylor:
the FASTEST runners (top sprint speed, quickest 90 ft) who get caught anyway — squandering an obvious speed advantage.

Josh Naylor 2025 — 24.4 ft/s, 1.9th speed percentile — ranks #5 of 357. Bobby Witt Jr. — 100th percentile — sits in the bottom tier every season tracked.

Pure Statcast · calendar-year data · no computer vision required

Generated 2026-05-30 · Companion: Variable_Glossary.pdf

Read This First — The Idea in 60 Seconds

One score, two poles, calendar-year data.

Stealing bases is treated here as a CONVERSION SKILL, not a speed contest.

The one number: Blueprint Conversion Score (BCS)

Every runner-season gets a single score. Positive = better than the running game expects given that runner's speed. Negative = worse.

What pushes a runner UP

- Covering a lot of ground from the pitcher's first move to release (a big 'jump') — ESPECIALLY if they are slow.
- Actually converting those jumps into stolen bases (high SB%).

What pushes a runner DOWN

- Being fast and getting caught anyway (wasting the speed).
- Repeatedly running into outs (many caught-stealings).
- Covering ground but still failing — squandering the jump too.

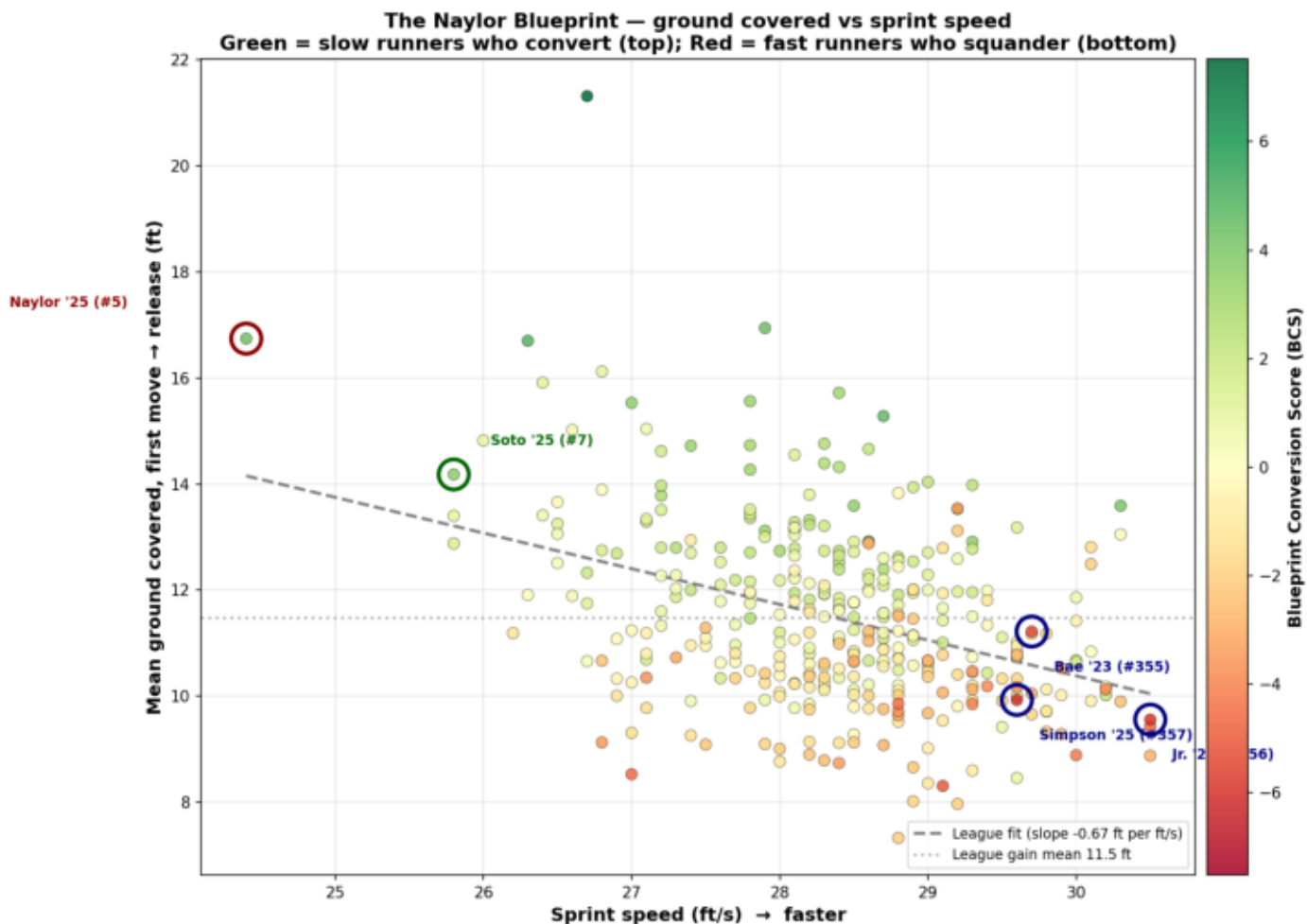
Why this matters

- 357 qualified runner-seasons (2023-2025), ≥ 10 tracked attempts each.
- The model surfaces 22 'blueprint' runners (slow + converts) and flags 96 'squander' runners (fast + fails).
- Josh Naylor 2025 ranks #5; Juan Soto 2025 ranks #7.

The punchline

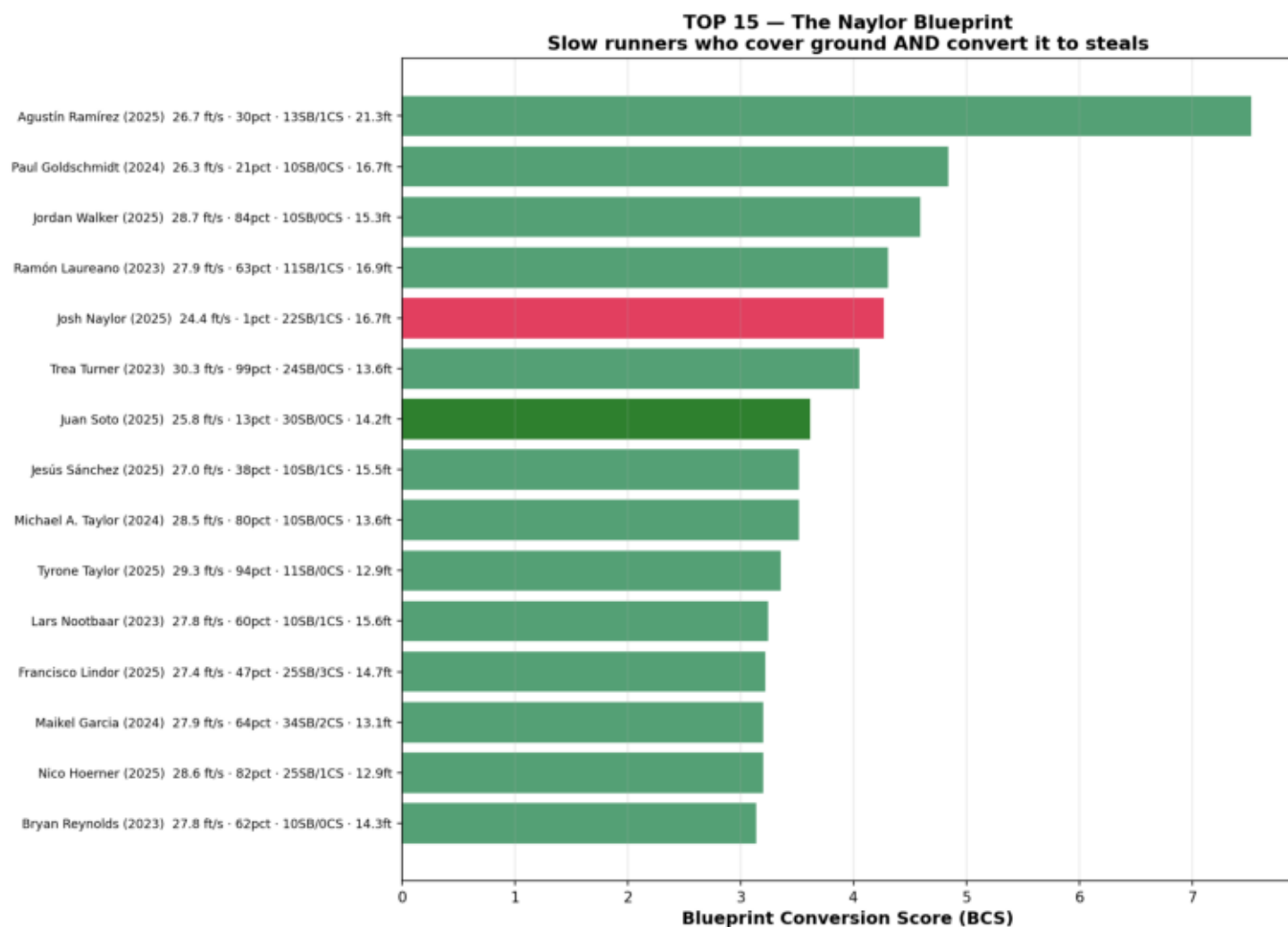
Speed barely buys ground. The runners who win the steal game on TIMING — not wheels — are a real, measurable, and rare archetype. Naylor is its gold standard; the fastest runners who keep getting caught are its mirror.

1 · The Big Picture



Each dot is a runner-season. X = sprint speed (right is faster); Y = mean ground covered from first move to release. Color = BCS (green is good, red is bad). The dashed league fit is nearly FLAT — speed barely buys ground, which is the whole point. Naylor (#5) and Soto (#7) sit high on the LEFT (slow but cover ground); the fastest squanderers sit low on the RIGHT.

2 · The Blueprint — Top 15



The runners who convert ground into steals relative to their speed. Naylor (crimson) and Soto (dark green) are highlighted. Most are slow to average runners with elite jumps and near-perfect success.

2 · The Blueprint — Top 15 (detail)

Slow runners who cover ground AND convert it.

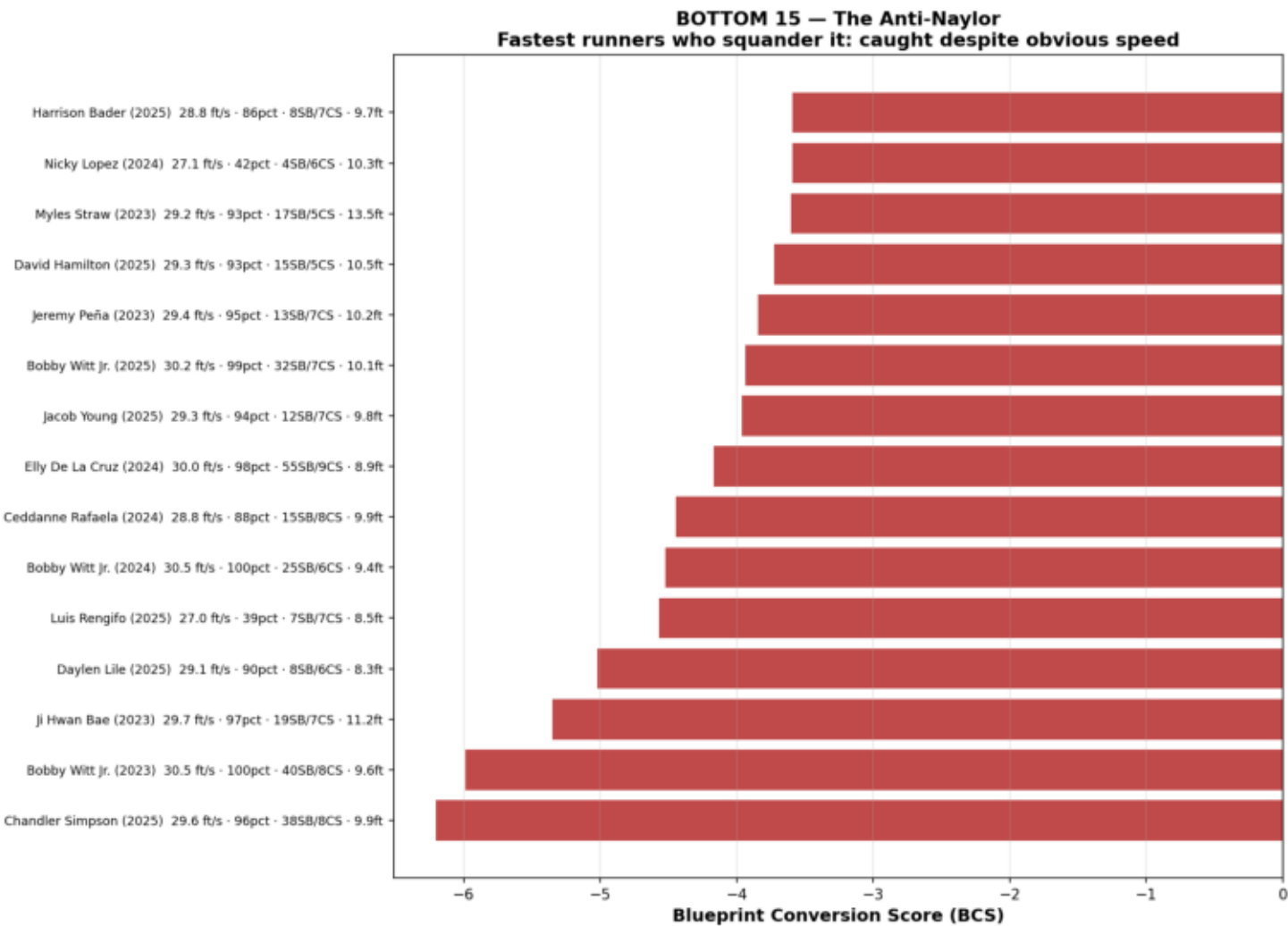
spd = sprint ft/s (percentile) · 90ft = home-to-first sec · gain = ground covered (ft)

# 1	Agustín Ramírez	2025	spd 26.7(30)	90ft 4.78	13/1= 93%	gain 21.3	BCS +7.52
# 2	Paul Goldschmidt	2024	spd 26.3(21)	90ft 4.58	10/0=100%	gain 16.7	BCS +4.84
# 3	Jordan Walker	2025	spd 28.7(84)	90ft 4.40	10/0=100%	gain 15.3	BCS +4.59
# 4	Ramón Laureano	2023	spd 27.9(63)	90ft 4.40	11/1= 92%	gain 16.9	BCS +4.31
# 5	Josh Naylor	2025	spd 24.4(1)	90ft 4.86	22/1= 96%	gain 16.7	BCS +4.27
# 6	Trea Turner	2023	spd 30.3(99)	90ft 4.14	24/0=100%	gain 13.6	BCS +4.05
# 7	Juan Soto	2025	spd 25.8(13)	90ft 4.58	30/0=100%	gain 14.2	BCS +3.61
# 8	Jesús Sánchez	2025	spd 27.0(38)	90ft 4.48	10/1= 91%	gain 15.5	BCS +3.51
# 9	Michael A. Taylor	2024	spd 28.5(80)	90ft 4.30	10/0=100%	gain 13.6	BCS +3.51
# 10	Tyrone Taylor	2025	spd 29.3(94)	90ft 4.38	11/0=100%	gain 12.9	BCS +3.35
# 11	Lars Nootbaar	2023	spd 27.8(60)	90ft 4.30	10/1= 91%	gain 15.6	BCS +3.24
# 12	Francisco Lindor	2025	spd 27.4(47)	90ft 4.40	25/3= 89%	gain 14.7	BCS +3.22
# 13	Maikel Garcia	2024	spd 27.9(64)	90ft 4.44	34/2= 94%	gain 13.1	BCS +3.20
# 14	Nico Hoerner	2025	spd 28.6(82)	90ft 4.34	25/1= 96%	gain 12.9	BCS +3.20
# 15	Bryan Reynolds	2023	spd 27.8(62)	90ft 4.44	10/0=100%	gain 14.3	BCS +3.14

How to read a line

- Naylor: 24.4 ft/s (1.9th pctl), slow 90 ft (4.86s), yet covers 16.7 ft and steals at 96% — a top-5 score on a bottom-2% motor.
- The blueprint is NOT 'be fast'. It is 'get a great jump and cash it in'.

3 · The Anti-Naylor — Bottom 15



The fastest runners who squander it. Every bar is red (negative BCS). These are top-percentile sprinters with the quickest 90-ft times who nonetheless rack up caught-stealings — the profile Naylor would kill for, wasted.

3 · The Anti-Naylor — Bottom 15 (detail)

Fastest runners, highest caught-stealing — penalized hardest.

The worst conversion scores on the board — fast, yet caught.

#357	Chandler Simpson	2025	spd 29.6(96)	90ft 3.97	38/8= 83%	gain 9.9	BCS -6.20
#356	Bobby Witt Jr.	2023	spd 30.5(100)	90ft 4.12	40/8= 83%	gain 9.6	BCS -5.99
#355	Ji Hwan Bae	2023	spd 29.7(97)	90ft 4.05	19/7= 73%	gain 11.2	BCS -5.35
#354	Daylen Lile	2025	spd 29.1(90)	90ft 4.23	8/6= 57%	gain 8.3	BCS -5.02
#353	Luis Rengifo	2025	spd 27.0(39)	90ft 4.36	7/7= 50%	gain 8.5	BCS -4.57
#352	Bobby Witt Jr.	2024	spd 30.5(100)	90ft 4.10	25/6= 81%	gain 9.4	BCS -4.53
#351	Ceddanne Rafaela	2024	spd 28.8(88)	90ft 4.17	15/8= 65%	gain 9.9	BCS -4.44
#350	Elly De La Cruz	2024	spd 30.0(98)	90ft 4.21	55/9= 86%	gain 8.9	BCS -4.17
#349	Jacob Young	2025	spd 29.3(94)	90ft 4.25	12/7= 63%	gain 9.8	BCS -3.96
#348	Bobby Witt Jr.	2025	spd 30.2(99)	90ft 4.15	32/7= 82%	gain 10.1	BCS -3.93
#347	Jeremy Peña	2023	spd 29.4(95)	90ft 4.28	13/7= 65%	gain 10.2	BCS -3.84
#346	David Hamilton	2025	spd 29.3(93)	90ft 4.02	15/5= 75%	gain 10.5	BCS -3.72
#345	Myles Straw	2023	spd 29.2(93)	90ft 4.14	17/5= 77%	gain 13.5	BCS -3.60
#344	Nicky Lopez	2024	spd 27.1(42)	90ft 4.31	4/6= 40%	gain 10.3	BCS -3.59
#343	Harrison Bader	2025	spd 28.8(86)	90ft 4.31	8/7= 53%	gain 9.7	BCS -3.59

The signature

- Chandler Simpson 2025 anchors the bottom (#357): 29.6 ft/s (96th pctl), 38/8, only 9.9 ft of ground.
- Bobby Witt Jr. — a 100th-percentile sprinter — lands in the bottom tier in EVERY season tracked. Obvious speed, squandered.

4 · Josh Naylor — The Archetype

A top-5 score on a bottom-2% motor.

The profile

- Sprint speed: 24.4 ft/s (1.9th percentile — bottom 2%)
- Home-to-1B: 4.86 s (slow)
- Ground covered: 16.74 ft (among the most in MLB)
- Steal record: 22/23 = 95.7%
- BCS: +4.27 → rank #5 of 357

Why he ranks so high

- Gain residual (ground vs. what speed predicts): +1.70 SD
- Success residual (steals vs. what speed predicts): +2.03 SD
- Squander penalty: -0.54 (the floor — too slow to squander a speed edge)

The rarity

Being THIS slow and covering THIS much ground is almost a contradiction. Treating the two as independent, the joint probability is on the order of 1 in tens of millions of runner-seasons. Naylor is a genuine outlier — the model surfaces him near the very top BECAUSE he is slow, not in spite of it.

Juan Soto — the same skill, more speed

- 2023: 26.8 ft/s (34pct), 6/10=60%, gain 16.1 ft, BCS +0.92 (#116)
- 2024: 26.8 ft/s (32pct), 6/10=60%, gain 13.9 ft, BCS +0.10 (#185)
- 2025: 25.8 ft/s (14pct), 30/30=100%, gain 14.2 ft, BCS +3.61 (#7)

5 · How the Score Works (Plain English)

Two rewards, one penalty — combined into a single number.

BCS adds two rewards and subtracts one penalty. All three are z-scores (standard deviations) across the qualified field, so they combine cleanly.

$$\text{BCS} = \text{success_resid_z} + \text{gain_resid_z} - \text{squander_z}$$

1) success_resid_z (reward)

- Start with a Beta-Binomial estimate of the runner's true steal rate, shrunk toward the league rate so a 4-for-4 cameo doesn't outrank a 30-for-31 season.
- Subtract what that runner's SPEED predicts. Positive = converts more often than a runner that fast 'should'.

2) gain_resid_z (reward)

- How much ground the runner covers from first move to release, minus what speed predicts. Positive = a big jump for how slow they are.

3) squander_z (penalty)

$$\text{raw} = \text{caught-stealings} \times \max(\text{speed}, 0) \times (1 + \max(\text{jump}, 0))$$

- ONLY fast runners can be penalized — you can't squander a speed advantage you don't have, so slow runners rest at the floor.
- The $(1 + \text{jump})$ factor is the 'failed to capitalize' clause: covering ground and STILL getting caught is punished harder.

The empirical-Bayes idea

Both rewards are shrunk toward league behavior when the sample is small, so the leaderboard rewards repeatable skill, not small-sample luck.

6 · Data Basis & Honest Limitations

What's measured, what's assumed, what's out of reach.

Annual (calendar-year) data — no career aggregates

Every input is bounded to a single season; one row = one runner-season.

- Stolen bases / caught stealing — MLB StatsAPI, season=Y, regular season
- Sprint speed & home-to-1B (90ft) — Baseball Savant, min=max=Y
- Per-attempt lead & ground covered — Baseball Savant, season_start=end=Y

A runner in 2023-25 contributes up to three independent rows.

On the speed metrics

Statcast publishes sprint speed (peak velocity) and home-to-1B (the 90-ft burst, a stand-in for acceleration). It does NOT publish per-player acceleration or jerk, so the speed composite uses sprint speed + 90-ft time.

What BCS does and does not claim

- It DOES rank conversion skill relative to a runner's own speed.
- It does NOT say a slow runner is 'better' than a fast one in a vacuum — it says who BEATS their own expectation, and who wastes their tools.
- Tracked attempts only (Savant lead tracking, 2023+). Volume-qualified at ≥ 10 tracked attempts to keep the residuals stable.

Why no computer vision

An earlier pipeline measured pitcher delivery time from video. It added NO predictive lift over the native Statcast lead metrics (leave-one-out AUC 0.838 without it vs. 0.822 with it). The honest answer was already in the Statcast data — so this model is pure Statcast.